

1(a)(i)	<i>direction</i> (drop of coloured liquid will move) left / AW and <i>explanation</i> (oxygen used in respiration) decreases / AW, volume / pressure (of gas in respirometer) ;	1 mark as prose allow (move) towards, <i>O. asellus</i> / test-tube / AW ignore move towards respirometer if nothing on answer line, check diagram and allow correct arrow / annotations on Figure 1.2 ignore <i>ref. to</i> capillary action ignore volume / pressure is low (needs to be lower) A 'a negative pressure is created'
1(a)(ii)	<i>idea of</i> replace (live) <i>O. asellus</i> with: sterile / inert / glass / plastic / AW, beads ; or dead / AW, <i>O. asellus</i>	1 allow woodlouse / woodlice / <i>asellus</i> / invertebrate / animal / organism as AW for <i>O. asellus</i> (shown in Figure 1.2) ignore respirometer with no <i>O. asellus</i> R if change another factor in addition to <i>O. asellus</i>

Question	Answer	Marks	Guidance
1(a)(iii)	any six from: 1 use a minimum of five stated temperatures ; 2 suitable <u>method</u> to maintain a constant temperature ; 3 use, same / stated, mass of, sodium hydroxide / NaOH (pellets) ; 4 use, same / stated, number / amount / mass, of <i>O. asellus</i> or use, same / stated, age of <i>O. asellus</i> ;	6	ignore ref. to control experiment allow woodlouse / woodlice / <i>asellus</i> / invertebrate / animal / organism as AW for <i>O. asellus</i> 1 temperatures must include 5 °C and 35 °C, but do not need to be evenly spaced must see units at least once ignore temperatures outside the range 2 A water-bath / incubator ignore temperature-controlled room / air conditioning / heater / fridge A ref. to use thermometer and ice / warm water (in a beaker) R Bunsen burner mp3, 4 & 6 allow standardised / controlled / fixed / constant as AW for same ignore known 3 ignore amount / number, of NaOH pellets ignore volume / concentration allow any value for mass if stated 4 ignore size / length / volume / concentration, of <i>O. asellus</i> ignore same species / type / sex (of <i>O. asellus</i>) allow any value for number / mass if stated ignore <i>O. asellus</i> (collected / bought) from same, environment / population / supplier ignore same, food source of <i>O. asellus</i> (before investigation)

Question	Answer	Marks	Guidance									
1(a)(iii)	5 <u>method</u> to, equilibrate / acclimatise, <i>O. asellus</i> ; 6 at each temperature, measure the distance (moved by the drop of coloured liquid), in a, set / stated, time or at each temperature, measure the time (for the drop of coloured liquid), to travel a, set / stated, distance ; 7 for each temperature, take at least three measurements ; 8 safety comment with hazard and risk and precaution ; <table><tr><th>hazard</th><th>risk</th><th>precaution</th></tr><tr><td>sodium hydroxide (pellets)</td><td>corrosive / caustic / damage to eyes</td><td>(wear) gloves / goggles / safety glasses / PPE</td></tr><tr><td><i>O. asellus</i> / woodlouse</td><td>allergy / biohazard / pathogens / infection</td><td>(wear) gloves / PPE</td></tr></table>	hazard	risk	precaution	sodium hydroxide (pellets)	corrosive / caustic / damage to eyes	(wear) gloves / goggles / safety glasses / PPE	<i>O. asellus</i> / woodlouse	allergy / biohazard / pathogens / infection	(wear) gloves / PPE		5 A leave <i>O. asellus</i> (in respirometer), in water-bath / at a set temperature, for, set / stated, time if stated, minimum time 5 minutes 6 allow note / record as AW for measure ignore calculate as AW for measure allow length / mm / cm as AW for distance ignore known time (in question stem) allow 'how far' as AW for distance allow 'how long' as AW for time ignore <i>ref. to</i> measuring without sodium hydroxide allow measuring the initial and final distance allow any value for distance / time if stated do not allow measuring distance and time if neither is set / stated 7 ignore calculate a mean A three replicates / repeat 2 times, at each temperature 8 ignore low / medium / high, risk R no risk ignore heat / temperature / hot water as hazard ignore irritant / toxic / burns as risk for sodium hydroxide ignore irritant / toxic as risk for <i>O. asellus</i>
hazard	risk	precaution										
sodium hydroxide (pellets)	corrosive / caustic / damage to eyes	(wear) gloves / goggles / safety glasses / PPE										
<i>O. asellus</i> / woodlouse	allergy / biohazard / pathogens / infection	(wear) gloves / PPE										

Question	Answer	Marks	Guidance
1(a)(iv)	distance (in mm) and multiplied by, πr^2 / area (of glass tube cross-section) and divided by time (in minutes) ;	1	allow d / h / l for distance allow t for time allow over time / multiply by 1/t, as AW for divide by time ignore per as AW for divided (in question stem) ignore units if stated A $d \pi r^2 / t$ r = radius of glass tube allow $\pi d^2 / 4 = \pi r^2$ (where d = diameter)
1(b)(i)	(different / type of) respiratory substrate (solutions) ;	1	A (20%) glucose and (20%) maltose (solutions) A (different / type of) sugar (solutions) R concentration of, glucose / maltose / sugar (solutions)
1(b)(ii)	add 4 cm ³ , (50%) stock (solution of glucose) to, 6 cm ³ , distilled water ;	1	allow ml as AW for cm³ allow (50%) glucose (solution) as AW for stock solution allow deionised / pure, water as AW for distilled water ignore water unqualified ignore $C_1V_1 = C_2V_2$ A use 4 cm³ stock solution and top up to 10 cm³, with distilled water

Question	Answer	Marks	Guidance
1(b)(iii)	any two from: 1 concentration of yeast / mass of yeast (in suspension) ; 2 species / variety / type / age / AW, of yeast ; 3 temperature ; 4 pH ; 5 size of (small) test-tube ;	2	ignore volume of yeast suspension ignore volume / concentration, of, respiratory substrate / glucose / maltose / sugar, solution ignore time ignore light intensity ignore carbon dioxide / oxygen, concentration R type of respiratory substrate R volume of gas 3 R temperature that (solutions / AW) were <u>boiled</u> at
1(c)(i)	<i>type of data:</i> nominal / discontinuous ;	1	ignore categoric / qualitative / independent (<i>as these are types of variable</i>) R quantitative / continuous / interval / ordinal / discrete / dependent

Question	Answer	Marks	Guidance						
1(c)(ii)	(mean) volume (of gas collected after 30 minutes), is, greater / AW, with, glucose (than with maltose) / ora and data quote (from Figure 1.5) ;	1	allow <u>rate</u> of, fermentation / (anaerobic) respiration, as AW for volume (of gas) ignore (glucose is) better / AW data quotes: candidates should give data quotes for glucose and maltose or the calculated difference. need units at least once allow ml as AW for cm³ <table><tr><th>respiratory substrate</th><th>volume of gas / cm³</th></tr><tr><td>glucose maltose</td><td>1.85 to 1.90 (e.g. 1.875) 1.45</td></tr><tr><td>calculated difference</td><td>0.40 to 0.45 (e.g. 0.425)</td></tr></table>	respiratory substrate	volume of gas / cm ³	glucose maltose	1.85 to 1.90 (e.g. 1.875) 1.45	calculated difference	0.40 to 0.45 (e.g. 0.425)
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1(c)(iii)	any two from: 1 maltose, is a <u>disaccharide</u> / is not a <u>monosaccharide</u>, and needs to be hydrolysed (into glucose) or glucose, is a <u>monosaccharide</u> / is not a <u>disaccharide</u>, and does not need to be hydrolysed ; 2 <i>idea that</i> only glucose enters, <u>glycolysis</u> ; ora 3 <i>idea that</i> entry of glucose into yeast cells is faster ;	2	ignore easier / more efficient ignore harder / more difficult ignore spelling of underlined words if it is clear what is meant 1 allow broken down / digested / converted (to glucose), as AW for hydrolysed ignore glucose is, broken down / AW ignore glucose is, easier / AW, to break down ignore maltose is, harder / AW, to break down 2 A maltose cannot enter <u>glycolysis</u> (unless it has been broken down) ignore respiration unqualified 3 A more glucose transporters available in yeast cells (than maltose transporters) A <i>idea that</i> gene expression for maltose, transporter / enzyme / maltase / AW, takes time